

What Are the Qualities of Good Malware Datasets?

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Agenda

- Project Description
- Project Objective and Goals
- Investigator Biographies
- Research Plan summary
- Project Deliverables
- Progress to date
- Conclusions to date
- Issues/Challenges
- Bibliography
- Acronyms
- Terms/Definitions

Project Description

Project Goal: Evaluating Malware Datasets

1. Research Question :

- What Are the Qualities of Good Malware Datasets?

2. Focus Areas:

- **Limitations and Biases:** Identify issues within current datasets.
- **Key Characteristics:** Define what constitutes a "good" malware dataset.

3. Outcome:

- Establish guidelines for creating more realistic and comprehensive datasets.
- Enhance the effectiveness of malware detection and analysis tools.

Project Objectives / Goals

1. Establish Best Practices:

- Develop guidelines for **high-quality malware dataset construction**.

2. Improve Dataset Generalization:

- Ensure datasets are **applicable to real-world threats**.

3. Enhance Detection Tools:

- **Boost the effectiveness** of malware detection and analysis.

4. Cross-Disciplinary Insights:

- Leverage recommendations from other fields to **broaden dataset design strategies**.

Investigator Biographies

- **Mohammad Rafieian:** Ph.D. candidate focused on **advancing software security** through **machine learning-driven vulnerability detection**.
- **Tharun Mulka:** Cybersecurity Ph.D. student specializing in **cryptography** and **innovative defenses against USB-based attacks**.
- **Sahithi Reddy Mopuru:** Master's student excelling in **data science and AI**, with a track record in **healthcare-focused machine learning research**.
- **Sri Kruthi Madugula:** Data Science specialist with strengths in **big data analytics** and **developing predictive models for cybersecurity**.
- **Pranathi Munnangi:** Master's student integrating **AI, cloud computing**, and **cybersecurity**, dedicated to **enhancing safety in intelligent systems**.

Research Plan Summary

Research Objective:

- Identify key qualities, recommendations, and limitations of existing malware datasets.
- Propose best practices for future malware dataset creation.

Division of Tasks:

- **Literature Review:** Evaluated 30+ papers on malware datasets and related fields.
- **Feature Extraction:** Identified strengths, weaknesses, and recommendations from cybersecurity and other fields.
- **Cross-Industry Comparison:** Analyzed dataset recommendations from weather prediction, image processing, IoT, etc., for transferable qualities.

Research Plan Summary

Methodology:

- **Systematic Literature Review:** Gathered insights on dataset limitations and recommendations.
- **Comparative Analysis:** Explored how datasets in other fields deal with challenges (e.g., unpredictability, regular updates).
- **Synthesis of Findings:** Proposed recommendations for malware datasets based on cross-industry insights.

Expected Outcome:

- Develop comprehensive qualities and recommendations for future malware datasets.
- Highlight common shortcomings and propose ways to address them.
- Apply strategies from other fields to improve dataset robustness, scalability, and generalizability.

Project Deliverables

We have three deliverables:

1. **Spreadsheet:** A Google spreadsheet compiling the literature review, feature extraction, and cross-industry analysis. It includes findings from over 30 research papers, key characteristics for effective datasets, and techniques from other fields.
2. **Report:** A detailed document summarizing the findings, methodology, insights, and recommendations for improving malware datasets, based on the literature review and comparative analysis.
3. **Presentation:** Slides highlighting the project's key results, proposed best practices, and recommendations for future malware dataset development.

Progress to Date

Cybersecurity Dataset Challenges

Historical Coherence

- Training data must precede test data to avoid unrealistic performance.

Random Selection Bias

- Careful selection of samples is needed to prevent future data from skewing results.

Model Degradation Over Time

- As malware evolves, older training data may not effectively detect new threats.

Class Imbalance

- Commonly more benign than malicious samples; balancing techniques like SMOTE can help.

Static vs. Dynamic Analysis

- **Static:** Fast and scalable but struggles with obfuscation.
- **Dynamic:** Captures runtime behavior but is resource-intensive.

Compiler and Architecture Impacts on Malware Detection

Environment Dependence

- Tools optimized for specific OS may struggle on others (e.g., Windows tools on Linux).

Hardware Dependencies

- Some tools are tailored for specific architectures (Intel vs. AMD).

Compiler Variability

- Different compilers produce varied binary outputs, affecting analysis.

Compiler Flags

- Optimization flags influence binary characteristics and may introduce vulnerabilities.

Cross-Architecture Challenges

- Malware targeting one architecture may behave differently when recompiled for another.

GraphML and Image Processing Key Findings

Key Findings	GraphML	Image Processing
Scalability Challenges	Real-world graphs are larger; OGB provides datasets up to 100M nodes.	Datasets should represent realistic conditions.
Lack of Diversity	Structural attributes like node degrees and clustering coefficients are limited in most datasets.	Datasets must avoid biases and include diverse malware samples.
Evaluation Protocols	Standardized evaluation (fixed data splits, metrics) ensures consistent results.	Cross-dataset generalization is key—models must perform well across datasets.
Data Splitting Methods	OGB uses time-based or structure-based splits for more realistic scenarios.	Incorporating realistic negative samples ensures model accuracy and reduces overconfidence.
Handling Rare Events	Include rare but significant graph events.	Dataset should capture rare malware attacks, like APTs and zero-day exploits.

IoT and Weather Dataset Design Considerations

Key Findings	IoT and Networks	Atmospheric Sciences and Weather Prediction
Dataset Balance	Balancing legitimate and malicious traffic is critical for accurate intrusion detection.	Datasets must capture diverse weather patterns, including rare extreme events like hurricanes.
Feature Selection	Features like message types, QoS levels, and session management are key to detecting subtle attacks.	Selecting critical weather variables (e.g., temperature, wind speed) enhances model accuracy.
Rare Event Inclusion	Attack-specific features improve detection for rare events like APTs.	Include rare but significant weather events to improve predictive models for extreme scenarios.
Model Adaptability	Ensemble models (e.g., Random Forests, RNNs) showed good performance in intrusion detection.	Models need frequent updates to handle evolving climatic conditions and new data sources (e.g., satellite data).

Summary of Results / Conclusions to Date

Findings on Malware Datasets:

- Existing malware datasets are **too specialized** and lack **generalization** across different projects.
- Researchers tend to create **new datasets** for specific tasks, resulting in an **abundance** of non-interchangeable datasets.
- Our evaluation distilled the most **essential features** recommended by researchers for malware datasets.

Broadening Investigation:

- Explored other fields with **similar challenges** to malware analysis.
 - Example: Malware analysis and **weather prediction** both require **regular updates** and anticipation of **unseen features** for accurate model performance.

Summary of Results / Conclusions to Date

Fields Examined (30 Papers Total):

- **Cybersecurity**
- **Image Processing & Graph Neural Networks**
- **Compilers & Program Analysis**
- **IoT & Network Security**
- **Weather Prediction**

Outcome:

- By combining insights from these fields, we propose a **comprehensive perspective** on enhancing the generalization and applicability of malware datasets.

Issues / Challenges

Lack of Standardized Benchmarks

- The absence of standardized benchmarks for evaluating malware datasets hinders consistent comparison and assessment across different datasets.

Challenges in Adopting Cross-Disciplinary Best Practices

- Adapting best practices from other fields is difficult due to structural differences in data, limiting the transferability of methodologies.

Scalability Limitations

- Techniques effective for small datasets may not perform well with larger datasets, posing scalability challenges in empirical analysis.

Generalizability of Characteristics

- Determining which dataset characteristics can be generalized across different dataset types and use cases remains a complex task.

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Acronyms

IoT: Internet of Things

APT: Advanced Persistent Threat

OGB: Open Graph Benchmark

RNN: Recurrent Neural Network

SMOTE: Synthetic Minority Over-sampling Technique

Terms / Definitions

Malware: Malicious software designed to disrupt, damage, or gain unauthorized access to computer systems.

Dataset Bias: The presence of systematic errors or skewness in data collection, leading to a dataset that does not fully represent the diversity or complexity of real-world malware.

Dataset Quality: The criteria by which a dataset is evaluated, including factors such as accuracy, completeness, consistency, and relevance.

Real-world Malware: Malware samples that are observed in actual cyber-attacks or infections, as opposed to artificial or laboratory-created samples.

Data Diversity: The range and variety of malware samples included in a dataset to ensure a comprehensive representation of different malware types and behaviors.

Malware Detection: The process of identifying malware based on its signatures, behaviors, or characteristics within a system or dataset.

False Positives/Negatives: Incorrect results in malware detection, where benign data is identified as malware (false positive) or malware goes undetected (false negative).

A representative malware dataset is a diverse and comprehensive collection of malware samples that accurately reflects the variety and complexity of real-world threats, ensuring effective analysis and detection across different malware types.